

AcerbE™ Integrity Verification Specification

Cryptographic Fingerprint Format, Pipeline, and Schema

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1. SCOPE

This specification defines the AcerbE™ fingerprint format, metadata schema, embedding pipeline, tamper-guard behavior, and verification procedure. It is the authoritative reference for integration, audit, and forensic use.

2. FINGERPRINT FORMAT

2.1 Fingerprint Block Structure

Field	Size	Encoding	Description
MAGIC	4 bytes	0xFEDCBA98	Block identifier
VERSION	1 byte	0x01	Protocol version
LENGTH	4 bytes	Big-endian uint32	Encrypted payload length
ENCRYPTED PAYLOAD	Variable	AES-256-CBC	IV (16B) + ciphertext
HMAC	32 bytes	HMAC-SHA256	Integrity seal over header+payload

2.2 Payload Schema (JSON, pre-encryption)

Key	Type	Description
fid	UUID string	Fingerprint ID – unique per transaction
email	string	Buyer email address
txid	string	Transaction ID
ts	ISO 8601 string	Issuance timestamp (UTC)
ver	string	Engine version identifier (ACERBE-FP-1.0)

3. KEY DERIVATION

Each fingerprint uses a buyer-unique AES-256 key derived via PBKDF2-HMAC-SHA256.

Parameters:

Parameter	Value
Algorithm	PBKDF2-HMAC-SHA256
Iterations	200,000
Key Length	32 bytes (256-bit)
Salt	SHA-256(fingerprint_id + transaction_id)
Master Key	ACERBE_MASTER_KEY environment variable

4. EMBEDDING PIPELINE

Format	Layer 1 Method	Capacity
PNG / JPG / WEBP	LSB pixel injection (RGB channels)	~3 bits/pixel
PDF	XMP metadata + DocInfo dictionary	Unlimited (hex-encoded)
MP3 / WAV / FLAC	Trailing-byte injection (post-EOF)	Unlimited
ZIP	Hidden manifest + inner-file trailing bytes	Unlimited

5. TAMPER-GUARD BEHAVIOR

Every stamped output is wrapped in an encrypted tamper-guard container.

Field	Specification
Container format	MAGIC(4) + FP_ID(36) + IV(16) + AES-256-CBC ciphertext + HMAC-SHA256(32)
HMAC scope	Full container header + encrypted content
Tamper response	Any byte modification invalidates HMAC. Extraction produces zero-byte output.
File extension	.rpx (RaPaX™ protected file)

6. REGISTRY SCHEMA

Column	Type	Description
fingerprint_id	TEXT PK	UUID – unique per transaction
buyer_name	TEXT	Full name of purchaser

buyer_email	TEXT	Email of purchaser
transaction_id	TEXT	Operator-supplied transaction identifier
product_id	TEXT	Operator-supplied product identifier
product_name	TEXT	Human-readable product name
file_format	TEXT	Detected format: image/pdf/audio/zip
original_hash	TEXT	SHA-256 of original file pre-fingerprint
issued_at	TEXT	ISO 8601 UTC timestamp
hmac_seal	TEXT	HMAC-SHA256 seal of the full record